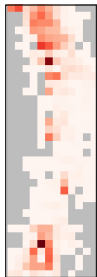


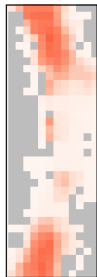


DINCAE

$|\hat{x} - x|$



$\hat{\sigma}$



4DVarNet aug. var. (ours)

$|\hat{x} - x|$



$\hat{\sigma}$



EnKF

$|\hat{x} - x|$



$\hat{\sigma}$

